Chapter 5. Ratio Estimation

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1 Auxiliary Information

So far the data were modeled as $y_i = \mu + \varepsilon_i$ with $\mathrm{E}(\varepsilon_i) = 0$, so $\mathrm{E}(y_i) = \mu$ for every unit. If an auxiliary variable x related to y is available for each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and knowledge of the population mean μ_x or total τ_x of x can be used to sharpen the estimate of μ_y or τ_y. This chapter uses a line through the origin (ratio estimation); Ch. 7 uses a general line (regression estimation).

Example 5.1: Sugar in a truckload of oranges

The wholesale price of oranges is based on the total sugar content τ_y of the load. The estimate $N\bar{y}$ needs the number of oranges N, which is unknown. But sugar content y is closely related to weight x, and the total weight τ_x of the truckload is easy to obtain. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\tau_x,$$

which does not require N.

Other ratios of interest: total monthly food budget to total monthly income per family; number of children to number of residents; average amount of a new drug needed to evoke a response relative to that of a standard drug.

2 Ratio Estimation under SRS

Let (x_i, y_i), $i = 1, \dots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - r x_i)^2$,

ratio:	$r = \bar{y}/\bar{x}$,	$\widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} \cdot \dfrac{1}{\mu_x^2}\, s_r^2$	(μ_x replaced by $\bar{x}$ if unknown);
total:	$\hat{\tau}_y = r\tau_x$,	$\widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \dfrac{1-f}{n} \dfrac{s_r^2}{\bar{x}^2}$;	
mean:	$\hat{\mu}_y = r\mu_x$,	$\widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} s_r^2$.	

The residuals $y_i - r x_i$ measure the departure from a line through the origin. If $y_i \approx r x_i$ for all units, s_r^2 is small and the ratio estimator is very precise.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^{N}(y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x} \cdot \frac{1}{n}\sum_{i=1}^{n}(y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^{N}(y_i - R x_i)^2 = \sum_{i=1}^{N}\left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum(x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = R x_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n} \cdot \frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is [1.0511, 1.0809].

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1 + \delta_y)(1 + \delta_x)^{-1} = R(1 + \delta_y)(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):
$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1 - f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x} - \mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x} - \mu_x)(\bar{y} - \mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1 - f}{n}S_x^2 - \frac{R}{\mu_x\mu_y}\frac{1 - f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ∎

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho\frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum(x_i - \mu_x)\,\beta(x_i - \mu_x)}{\frac{1}{N-1}\sum(x_i - \mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r,\bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r)-R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r,\bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r,\bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r)-R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the standard error whenever the coefficient of variation of $\bar{x}$ is small, which holds for moderately large n.

4 Ratio Estimation in Stratified Random Sampling

With strata $h = 1, \dots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h\, r_h\, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

   For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C\, \mu_x, \qquad r_C = \frac{\bar{y}_{\mathrm{st}}}{\bar{x}_{\mathrm{st}}}, \quad \bar{y}_{\mathrm{st}} = \sum_h w_h \bar{y}_h,\ \bar{x}_{\mathrm{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h}(y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$(N_h - 1)S_{h,S}^2 = \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h - 1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h \rho_h S_{xh} S_{yh}),$$

$$(N_h - 1)S_{h,C}^2 = \sum_i \left[(y_{hi} - \mu_{yh}) - R_C(x_{hi} - \mu_{xh})\right]^2$$
$$= (N_h - 1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C \rho_h S_{xh} S_{yh}).$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\text{st}} - R_C\bar{x}_{\text{st}} = \bar{d}_{\text{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C\mu_{xh}$.) Subtracting,

$$\begin{aligned} S^2_{h,C} - S^2_{h,S} &= (R_C^2 - R_h^2)S^2_{xh} - 2(R_C - R_h)\rho_h S_{xh}S_{yh} \\ &= (R_C - R_h)^2 S^2_{xh} + 2(R_C - R_h)\big(R_h S^2_{xh} - \rho_h S_{xh}S_{yh}\big). \end{aligned}$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\operatorname{Var}(\hat{\mu}_{yRC}) - \operatorname{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h}(R_C - R_h)^2 S^2_{xh} \;\geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But each r_h carries its own ratio bias, and these add up across strata. When the n_h are small the separate estimator can be badly biased; when the n_h are large (over about 50) the bias is negligible. With small strata the combined estimator is safer.

5 Ratio Estimator versus the Sample Mean

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\operatorname{RE}(\hat{\theta}_1, \hat{\theta}_2) = \operatorname{Var}(\hat{\theta}_1)/\operatorname{Var}(\hat{\theta}_2)$:

$$\operatorname{Var}(\bar{y}) = \frac{1-f}{n}S_y^2, \qquad \operatorname{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\operatorname{RE}(\bar{y}, \hat{\mu}_y) = \frac{\operatorname{Var}(\bar{y})}{\operatorname{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\operatorname{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2}\cdot\frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2}\cdot\frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\operatorname{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ■

The ratio estimator is more efficient than $\bar{y}$ when x and y are highly positively correlated; the estimated relative efficiency is meaningful only for large samples.